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import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from scipy import stats
import statsmodels.api as sm
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score

df = pd.read_csv("WA_Fn-UseC_-HR-Employee-Attrition.csv")

print("Dataset Loaded Successfully!")
print("Shape:", df.shape)

print("\nFirst 5 Rows:")
print(df.head())

print("\nDataset Information:")
print(df.info())

print("\nMissing Values:")
print(df.isnull().sum())

numerical_columns = [
    "Age",
    "MonthlyIncome",
    "YearsAtCompany",
    "YearsInCurrentRole",
    "JobSatisfaction"
]

print("\nMean:")
print(df[numerical_columns].mean())

print("\nMedian:")
print(df[numerical_columns].median())

print("\nMode:")
print(df[numerical_columns].mode().iloc[0])

print("\nVariance:")
print(df[numerical_columns].var())

print("\nStandard Deviation:")
print(df[numerical_columns].std())

print("\nComplete Statistical Summary:")
print(df[numerical_columns].describe())

correlation_matrix = df[numerical_columns].corr(method="pearson")

print("\nPearson Correlation Matrix:")
print(correlation_matrix)

plt.figure(figsize=(10, 7))

sns.heatmap(
    correlation_matrix,
    annot=True,
    cmap="coolwarm",
    fmt=".2f"
)

plt.title("Pearson Correlation Matrix")
plt.show()

left = df[df["Attrition"] == "Yes"]["MonthlyIncome"]
stayed = df[df["Attrition"] == "No"]["MonthlyIncome"]

t_statistic, p_value = stats.ttest_ind(
    left,
    stayed,
    equal_var=False
)
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)

print("\nIndependent Sample t-test")
print("T-statistic:", t_statistic)
print("P-value:", p_value)

alpha = 0.05

if p_value < alpha:
    print("Decision: Reject H0")
    print("Conclusion: Monthly income differs significantly between the two groups.")
else:
    print("Decision: Fail to reject H0")
    print("Conclusion: No significant difference was found.")

groups = []

for role in df["JobRole"].unique():
    group = df[df["JobRole"] == role]["JobSatisfaction"]
    groups.append(group)

f_statistic, anova_p_value = stats.f_oneway(*groups)

print("\nOne-Way ANOVA")
print("F-statistic:", f_statistic)
print("P-value:", anova_p_value)

if anova_p_value < alpha:
    print("Decision: Reject H0")
    print("Conclusion: Job satisfaction differs significantly among job roles.")
else:
    print("Decision: Fail to reject H0")
    print("Conclusion: No significant difference was found among job roles.")

X = df[["YearsAtCompany"]]
y = df["MonthlyIncome"]

model = LinearRegression()

model.fit(X, y)

y_pred = model.predict(X)

print("\nLinear Regression")
print("Intercept:", model.intercept_)
print("Coefficient:", model.coef_[0])

r2 = r2_score(y, y_pred)

print("R² Score:", r2)

plt.figure(figsize=(10, 6))

plt.scatter(
    df["YearsAtCompany"],
    df["MonthlyIncome"],
    alpha=0.5
)

plt.plot(
    df["YearsAtCompany"],
    y_pred
)

plt.xlabel("Years at Company")
plt.ylabel("Monthly Income")

plt.title(
    "Linear Regression: Years at Company vs Monthly Income"
)

plt.show()

X_sm = sm.add_constant(df["YearsAtCompany"])

regression_model = sm.OLS(
    df["MonthlyIncome"],
    X_sm

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).fit()

print("\nStatsmodels Regression Results:")
print(regression_model.summary())

print("\n95% Confidence Intervals:")
print(regression_model.conf_int())

print("\n=====")
print("FINAL CONCLUSION")
print("=====")

print("1. Descriptive statistics summarize the dataset.")
print("2. Pearson correlation identifies relationships between numerical variables.")
print("3. The t-test compares monthly income between employees who left and stayed.")
print("4. ANOVA compares job satisfaction across different job roles.")
print("5. Linear regression analyzes the relationship between years at company and monthly income.")
print("6. P-values help determine statistical significance.")
print("7.  $R^2$  indicates how much variation is explained by the regression model.")
```

Dataset Loaded Successfully!
Shape: (1470, 35)

First 5 Rows:

	Age	Attrition	BusinessTravel	DailyRate	Department	
0	41	Yes	Travel_Rarely	1102		Sales
1	49	No	Travel_Frequently	279	Research & Development	
2	37	Yes	Travel_Rarely	1373	Research & Development	
3	33	No	Travel_Frequently	1392	Research & Development	
4	27	No	Travel_Rarely	591	Research & Development	

	DistanceFromHome	Education	EducationField	EmployeeCount	EmployeeNumber	
0	1	2	Life Sciences	1	1	
1	8	1	Life Sciences	1	2	
2	2	2	Other	1	4	
3	3	4	Life Sciences	1	5	
4	2	1	Medical	1	7	

	RelationshipSatisfaction	StandardHours	StockOptionLevel	
0	...	1	80	0
1	...	4	80	1
2	...	2	80	0
3	...	3	80	0
4	...	4	80	1

	TotalWorkingYears	TrainingTimesLastYear	WorkLifeBalance	YearsAtCompany	
0	8	0	1	6	
1	10	3	3	10	
2	7	3	3	0	
3	8	3	3	8	
4	6	3	3	2	

	YearsInCurrentRole	YearsSinceLastPromotion	YearsWithCurrManager
0	4	0	5
1	7	1	7
2	0	0	0
3	7	3	0
4	2	2	2

[5 rows x 35 columns]

Dataset Information:

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 1470 entries, 0 to 1469

Data columns (total 35 columns):

#	Column	Non-Null Count	Dtype
0	Age	1470 non-null	int64
1	Attrition	1470 non-null	object
2	BusinessTravel	1470 non-null	object
3	DailyRate	1470 non-null	int64
4	Department	1470 non-null	object
5	DistanceFromHome	1470 non-null	int64
6	Education	1470 non-null	int64
7	EducationField	1470 non-null	object
8	EmployeeCount	1470 non-null	int64
9	EmployeeNumber	1470 non-null	int64
10	EnvironmentSatisfaction	1470 non-null	int64
11	Gender	1470 non-null	object
12	HourlyRate	1470 non-null	int64
13	JobInvolvement	1470 non-null	int64
14	JobLevel	1470 non-null	int64
15	JobRole	1470 non-null	object
16	JobSatisfaction	1470 non-null	int64
17	MaritalStatus	1470 non-null	object
18	MonthlyIncome	1470 non-null	int64
19	MonthlyRate	1470 non-null	int64
20	NumCompaniesWorked	1470 non-null	int64
21	Over18	1470 non-null	object
22	OverTime	1470 non-null	object
23	PercentSalaryHike	1470 non-null	int64
24	PerformanceRating	1470 non-null	int64
25	RelationshipSatisfaction	1470 non-null	int64
26	StandardHours	1470 non-null	int64
27	StockOptionLevel	1470 non-null	int64
28	TotalWorkingYears	1470 non-null	int64
29	TrainingTimesLastYear	1470 non-null	int64
30	WorkLifeBalance	1470 non-null	int64
31	YearsAtCompany	1470 non-null	int64
32	YearsInCurrentRole	1470 non-null	int64
33	YearsSinceLastPromotion	1470 non-null	int64
34	YearsWithCurrManager	1470 non-null	int64

dtypes: int64(26), object(9)
memory usage: 402.1+ KB
..

None

Missing Values:

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Age                0
Attrition           0
BusinessTravel      0
DailyRate           0
Department          0
DistanceFromHome    0
Education           0
EducationField       0
EmployeeCount        0
EmployeeNumber       0
EnvironmentSatisfaction 0
Gender              0
HourlyRate           0
JobInvolvement       0
JobLevel            0
JobRole             0
JobSatisfaction      0
MaritalStatus        0
MonthlyIncome        0
MonthlyRate          0
NumCompaniesWorked   0
Over18              0
OverTime             0
PercentSalaryHike     0
PerformanceRating     0
RelationshipSatisfaction 0
StandardHours        0
StockOptionLevel     0
TotalWorkingYears    0
TrainingTimesLastYear 0
WorkLifeBalance      0
YearsAtCompany       0
YearsInCurrentRole    0
YearsSinceLastPromotion 0
YearsWithCurrManager 0

```

dtype: int64

Mean:

```

Age                36.923810
MonthlyIncome      6502.931293
YearsAtCompany      7.008163
YearsInCurrentRole  4.229252
JobSatisfaction     2.728571

```

dtype: float64

Median:

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Age                36.0
MonthlyIncome      4919.0
YearsAtCompany      5.0
YearsInCurrentRole  3.0
JobSatisfaction     3.0

```

dtype: float64

Mode:

```

Age                35
MonthlyIncome      2342
YearsAtCompany      5
YearsInCurrentRole  2
JobSatisfaction     4

```

Name: 0, dtype: int64

Variance:

```

Age                8.345505e+01
MonthlyIncome      2.216486e+07
YearsAtCompany      3.753431e+01
YearsInCurrentRole  1.312712e+01
JobSatisfaction     1.216270e+00

```

dtype: float64

Standard Deviation:

```

Age                9.135373
MonthlyIncome      4707.956783
YearsAtCompany      6.126525
YearsInCurrentRole  3.623137
JobSatisfaction     1.102846

```

dtype: float64

Complete Statistical Summary:

	Age	MonthlyIncome	YearsAtCompany	YearsInCurrentRole	\
count	1470.000000	1470.000000	1470.000000	1470.000000	
mean	36.923810	6502.931293	7.008163	4.229252	
std	9.135373	4707.956783	6.126525	3.623137	